

RESEARCH ARTICLE

Amorphized and Oxidized Layered PtSe₂ as an All-in-One Electrocatalytic Platform for Hydrazine-Assisted Water Splitting

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ABSTRACT

Hydrazine-assisted water electrolysis is a promising energy-saving alternative to conventional water splitting; however, noble-metal catalysts suffer severe deactivation from nitrogen-intermediate poisoning during sustained anodic operation. Herein, amorphized and oxidized layered PtSe₂ is presented as an all-in-one bifunctional electrocatalytic platform solving this challenge. Plasma treatment induces simultaneous surface oxidation, amorphization, and Pt clusterization to form vertically integrated Pt–PtSe_xO_y heterostructures, as verified by transmission electron microscopy and x-ray photoelectron spectroscopy. Density functional theory and in situ Raman spectroscopy reveal a potential-dependent reaction network: N–H dehydrogenation dominates at low bias, while N–N cleavage becomes accessible at elevated potentials. Within this network, the amorphous PtSe_xO_y matrix acts as an efficient electron sink and proton-accepting environment. By rendering Pt clusters optimally electron-deficient, it cooperatively facilitates proton-coupled electron transfer during N–H activation and fundamentally suppresses the accumulation of strongly bound nitrogen intermediates, resulting in enhanced intrinsic hydrazine oxidation activity with increased turnover frequencies. Simultaneously, the formation of Pt clusters within this modulated coordination environment drastically boosts the cathodic hydrogen evolution reaction beyond commercial Pt/C benchmarks. Leveraging this synergistic bifunctionality, a symmetric two-electrode electrolyzer achieves highly energy-efficient overall water splitting, delivering long-term stability and demonstrating state-of-the-art performance among transition metal compound-based electrocatalysts.

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1 | Introduction

The global pursuit of carbon neutrality and sustainable energy systems has positioned renewable electricity-driven water electrolysis as a cornerstone technology for clean hydrogen production [1–6]. However, the overall efficiency of water splitting is constrained by the sluggish oxygen evolution reaction (OER) at the anode, which requires a high thermodynamic potential of 1.23 V versus the reversible hydrogen electrode (RHE) and involves a complex four-electron transfer process with a substantial kinetic barrier associated with O–O bond formation [7–11]. These intrinsic limitations result in large overpotentials for the overall electrocatalysis and impede practical hydrogen generation [9, 12–14]. To circumvent the bottleneck of OER, hybrid water electrolysis systems have been proposed, in which the anodic half-reaction is replaced by a more kinetically favorable oxidation process [15–17]. Among the various alternatives, the hydrazine oxidation reaction (HzOR), which converts hydrazine (N_2H_4) into gaseous nitrogen, has attracted significant attention due to its remarkably low theoretical potential of -0.33 V versus RHE [15, 18, 19]. This ultralow operating potential originates from the facile adsorption of hydrazine on catalyst surfaces via the lone-pair electrons of nitrogen atoms, enabling efficient electron transfer and bond activation [9, 20, 21]. HzOR produces only benign nitrogen gas and water as products, without generating carbon-containing byproducts, thereby offering substantial energy savings and environmental advantages for electrochemical water splitting [9, 15].

Mechanistically, HzOR can proceed through either a stepwise deprotonation pathway or an alternative route involving initial N–N bond cleavage followed by deprotonation [15, 21–26]. The stepwise N–H dehydrogenation pathway is preferentially operative at low anodic potentials, where successive proton–electron transfer steps dominate the reaction kinetics [15, 21, 22]. In contrast, the N–N bond cleavage pathway becomes kinetically accessible only at relatively higher anodic potentials, where enhanced electron withdrawal from adsorbed N_2H_4 sufficiently weakens the N–N bond, ultimately leading to its dissociation into two NH_2 intermediates [15, 21, 22]. Platinum-based catalysts are widely regarded as benchmark materials for HzOR owing to their efficient activation of N–H bonds and relatively low onset potentials [15, 26–28]. On Pt surfaces, hydrazine oxidation predominantly proceeds through a stepwise N–H dehydrogenation pathway, which enables high turnover frequencies at low current densities. However, under sustained anodic polarization and increasing reaction flux, Pt catalysts suffer from progressive activity decay, which is commonly attributed to the accumulation of strongly adsorbed nitrogen-containing intermediates and localized electronic instability at the metal surface [28–30]. Although N–N bond cleavage is in principle possible on noble-metal surfaces, this pathway is kinetically disfavored and does not make a significant contribution to HzOR on Pt under typical anodic operating conditions [15, 31]. Consequently, Pt catalysts face an inherent trade-off between fast intrinsic turnover and long-term operational stability. In addition to these intrinsic limitations, the high cost and scarcity of Pt further motivate strategies that maximize Pt utilization while minimizing noble-metal loading.

Transition metal dichalcogenides (TMDs) have attracted considerable interest as electrocatalysts owing to their earth abundance,

chemical stability, and high intrinsic activity of edge sites; however, their application in HzOR remains limited [32–39]. This limitation mainly originates from the scarcity of catalytically active sites on the electrochemically inert basal planes of most TMDs, together with inefficient charge extraction and sluggish interfacial charge-transfer kinetics under anodic potentials [13, 40–42]. Although conventional TMDs frequently exhibit n-type semiconducting behavior due to prevalent chalcogen vacancies and extrinsic doping effects, their unfavorable electronic structure under oxidative bias—characterized by limited hole availability and inefficient electron withdrawal—impedes the continuous, multi-electron oxidation and successive deprotonation steps required for efficient HzOR catalysis [13, 40, 43–46]. In contrast, layered platinum-based chalcogenides undergo thickness-dependent electronic transitions from semiconducting mono- or bilayers to semimetallic multilayer structures, resulting in markedly enhanced electrical conductivity and charge-transfer kinetics [47–50]. Furthermore, controlled introduction of chalcogen vacancies exposes undercoordinated Pt atoms and promotes the formation of Pt–O species, substantially increasing active-site density and enhancing oxidative catalysis by rendering Pt centers electron-deficient [51–54]. To further advance HzOR electrocatalysis, recent studies have focused on transition metal phosphide catalysts, enhancing HER and HzOR activities through intricate compositional and electronic-structure engineering. Specifically, they utilize Al-doped $\text{Ni}_2\text{P}/\text{NF}$ regulated by phosphorus vacancies [55], $\text{FeNi-CoP}/\text{NF}$ combining electronic modulation with lattice strain [56], and an FeCoP/NF system operating at the $\text{Fe}_2\text{P-Co}_2\text{P}$ interface [57]. While these previous works have made remarkable progress, they primarily rely on multi-step, wet-chemical synthesis routes and heteroatom doping.

2 | Results and Discussion

Here we report a PtSe_2 -based bifunctional electrocatalyst that achieves high activity and operational stability for HzOR-assisted water electrolysis. A single-step O_2 plasma treatment induces simultaneous surface oxidation, amorphization, and Pt clusterization on layered PtSe_2 . Structural and compositional characterizations, including transmission electron microscopy, Raman- and x-ray photoelectron spectroscopy confirm the formation of a vertically integrated heterogeneous architecture, featuring Pt clusters interfaced with an amorphous PtSe_xO_y matrix. Density functional theory and operando Raman spectroscopy reveal a potential-dependent reaction network for HzOR, where stepwise N–H dehydrogenation dominates at low potentials and N–N cleavage activates at higher potentials. Notably, the amorphous PtSe_xO_y domains function as efficient electron sinks, continuously redistributing excess charge away from the Pt active sites, while simultaneously acting as a proton-accepting environment. This dual functionality facilitates proton-coupled electron transfer and suppresses the accumulation of strongly bound nitrogen intermediates, maintaining higher turnover frequencies and anodic durability compared with pure Pt benchmarks. Concurrently, the modulated coordination environment of the Pt clusters enhances the cathodic hydrogen evolution reaction (HER). Utilizing this bifunctionality, the optimized heterostructure operates as a symmetric two-electrode electrolyzer for energy-efficient overall water splitting.

Figure 1 illustrates the conceptual design of the bifunctional platform alongside the morphological evolution of the PtSe₂ surface with O₂ plasma treatment. Specifically, the schematic in Figure 1a highlights the core concept: the plasma-induced surface reconstruction generates exposed Pt clusters that serve as active sites for both HzOR and HER. These clusters are intimately coupled with an electron-withdrawing and proton-accepting amorphous PtSe_xO_y domains that prevents the poisoning of nitrogen intermediates, thereby enabling efficient HzOR-assisted water splitting. To experimentally realize this architecture over large-area (> cm²) electrodes, multilayer PtSe₂ flakes mechanically exfoliated from bulk PtSe₂ crystals were transferred in a stepwise manner onto Au/SiO₂/Si substrates, resulting in an average surface coverage of ~55%. This stepwise transfer process ensured a high degree of sample-to-sample consistency, which was maintained before and after subsequent O₂ plasma treatment, as verified by optical microscopy (Figure 1b, Experimental section, Figures S1–S3). Statistical analysis of flake thickness and lateral size distributions, obtained by atomic force microscopy (AFM), revealed that the transferred flakes exhibit an average thickness of ~28 nm (corresponding to ~35 PtSe₂ layers) and an average lateral dimension of ~0.81 μm; these structural dimensions were largely preserved after plasma treatment (Figures 1c and S4). AFM cross-sectional profiles of flakes with different lateral sizes—approximately ~0.76 μm (average), ~1.28 μm (above average), and ~0.28 μm (below average)—demonstrate that the initially flat PtSe₂ surfaces undergo a pronounced transformation into three-dimensionally (3D) corrugated nanostructures upon plasma exposure (Figure 1d,e and Table S1). Notably, the root-mean-square (RMS) surface roughness increases by approximately 2.5-fold, regardless of flake size. This size-independent increase in surface roughness suggests that the extent of plasma-induced selenium vacancy formation and the accompanying localized surface amorphization are consistent regardless of lateral dimensions. Accordingly, the observed morphological evolution is governed primarily by the plasma–surface interaction rather than by flake size.

To spatially resolve the plasma-induced surface reconstruction of PtSe₂ down to the atomic scale, transmission electron microscopy (TEM) analyses were performed (Figure 2). As-transferred flakes exhibit largely featureless and atomically flat surfaces in low-magnification TEM images (Figure 2a), consistent with the AFM topography shown in Figure 1d. High-resolution TEM (HRTEM) images reveal well-defined lattice fringes, indicative of a crystalline 1T-PtSe₂ structure, in agreement with previous reports [48, 58]. Energy-dispersive x-ray spectroscopy (EDX) mapping confirms a stoichiometric PtSe₂ composition, with only negligible oxygen signals originating from native oxide species localized at the outer edges of the flakes (Figure S5). Intensity line profiles of lattice fringes extracted from the corresponding HRTEM images further verify the lattice spacings of (110), (010), and (100) planes for single-crystalline PtSe₂ (c-PtSe₂), in good agreement with their fast Fourier transform (FFT) and selected-area electron diffraction (SAED) patterns (Figure 2b) [48, 58]. Given the multilayered nature of the flakes, HRTEM analyses were performed primarily at the flake edges; this approach enables reliable identification of surface crystallinity—including phase, orientation, and local disorder—because the edges of layered TMDs exist predominantly as monolayers, allowing atomic-scale interrogation [37]. Following O₂ plasma treatment,

pronounced surface corrugations become apparent, as evidenced by contrast variations across the flakes in low-magnification TEM images (Figure 2c), consistent with AFM measurements. Closer examination of HRTEM images and lattice fringe analyses reveals a dramatic surface reconstruction accompanied by the formation of heterogeneous nanoscale domains comprising: (1) clusters of polycrystalline Pt; (2) amorphous PtSe_xO_y regions exhibiting local structural disorder; and (3) residual domains of crystalline PtSe₂. In general, accumulated chalcogen vacancy concentrations in Pt-based TMDs are known to induce local Pt clusterization at the surface [58, 59], consistent with the structural motifs observed here.

This plasma-induced surface transformation is further corroborated by the corresponding FFT patterns, which display two characteristic diffraction rings associated with the (111) and (200) planes of polycrystalline Pt [58, 60], along with a diffuse amorphous halo indicative of locally disordered PtSe_xO_y domains, coexisting with discrete diffraction spots indexed to c-PtSe₂ (Figures 2d and S6). These additional diffraction features are mostly negligible in the SAED patterns, suggesting that the plasma-induced reconstruction is likely confined to the outermost surface layers and does not fully propagate throughout the multilayer flakes. When amorphization is confined to only a few outermost layers of mechanically exfoliated Pt-based TMDs while the underlying layers remain crystalline, the resulting amorphous diffraction features are often weak or undetectable in SAED patterns; notably, such surface-limited amorphization is frequently accompanied by the formation of Pt clusters at the surface [58, 61]. Consistent with the structural evolution, EDX compositional mapping reveals an increased oxygen concentration distributed across the flake surfaces after plasma treatment, which indicates that selenium vacancy formation and oxygen incorporation occur concurrently, leading to the formation of amorphous PtSe_xO_y domains accompanied by local Pt clustering (Figures 2e and S6).

We performed a series of spectroscopic analyses to understand the chemical and electronic evolution of PtSe₂ induced by O₂ plasma treatment (Figure 3). X-ray photoelectron spectroscopy (XPS) reveals pronounced chemical-state modulation of PtSe₂ that is consistent with the plasma-induced surface reconstruction analyzed by TEM. Owing to the relatively large x-ray beam spot size (Experimental section), the XPS spectra represent a statistical average over multiple PtSe₂ flakes and surface regions, thereby providing global information on the overall chemical evolution induced by plasma treatment. In the Pt 4f core-level spectrum, pristine PtSe₂ exhibits only the characteristic Pt–Se doublet at 73.1 and 76.4 eV (Figure 3a). After O₂ plasma treatment at 180 W, two additional spectral components emerge, which are assigned to metallic Pt clusters (Pt_n, where n denotes the number of constituent Pt atoms, as defined in the theoretical section) at 72.0 and 75.4 eV, and to oxidized PtO₂ species at 74.6 and 77.9 eV, respectively [59, 62]. Importantly, the emergence of the Pt_n signal directly evidences the formation of metallic Pt clusters at the surface, which originates from plasma-induced Pt–Se bond cleavage followed by selenium oxidation and volatilization as SeO₂, leading to selenium vacancy accumulation and subsequent Pt precipitation [54, 58, 59, 63]. Concomitantly, the Se 3d spectrum shows the emergence of additional peaks attributable to SeO₂

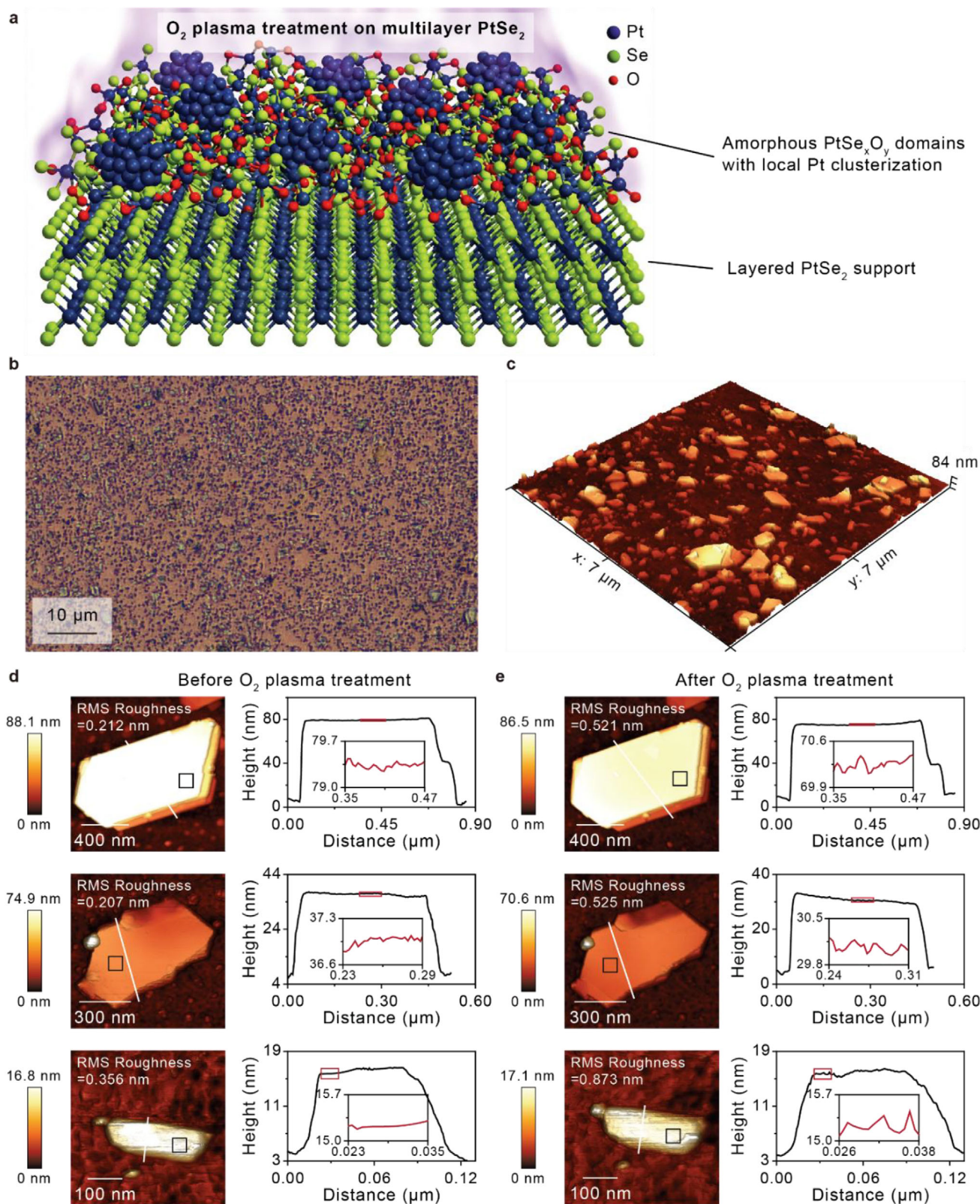


FIGURE 1 | Conceptual design and morphological evolution of the bifunctional $PtSe_2$ platform. (a) Schematic illustration of O_2 -plasma-treated multilayer $PtSe_2$, where the surface is transformed into amorphous $PtSe_xO_y$ domains with locally clustered Pt species, while the underlying layered $PtSe_2$ is retained as a support. (b) Optical microscope image of $PtSe_2$ flakes after O_2 plasma treatment. (c) Atomic force microscopy (AFM) topography of representative O_2 -plasma-treated $PtSe_2$ flakes. (d, e) AFM images of the same $PtSe_2$ flakes with different lateral dimensions before (d) and after (e) O_2 plasma treatment. Black dashed boxes indicate the regions used to calculate the root-mean-square (RMS) surface roughness, and white lines mark the locations from which height profiles were extracted.

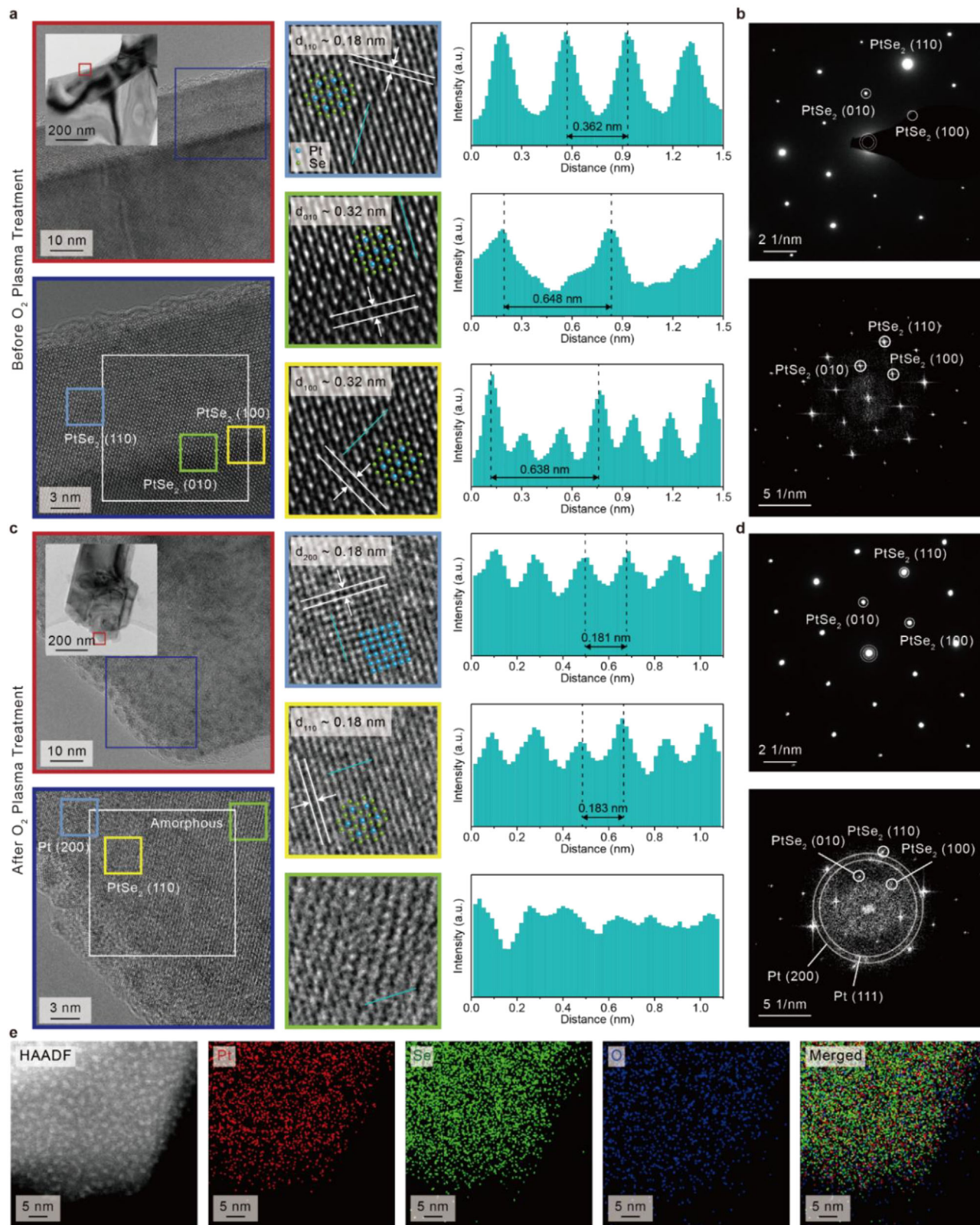


FIGURE 2 | Surface reconstruction of layered PtSe₂ revealed by TEM after O₂ plasma treatment. (a) Transmission electron microscopy (TEM) images of pristine PtSe₂ flakes, showing flat surfaces and well-preserved layered structures. (b) Selected-area electron diffraction (SAED) patterns and fast Fourier transform (FFT) images of pristine PtSe₂, confirming its crystalline 1T phase. (c) TEM images of PtSe₂ after O₂ plasma treatment, revealing pronounced surface corrugation and heterogeneous contrast. (d) Corresponding SAED and FFT patterns after plasma treatment, showing the coexistence of crystalline PtSe₂ diffraction spots, polycrystalline Pt rings, and diffuse amorphous features. (e) Energy-dispersive X-ray spectroscopy (EDX) elemental maps of PtSe₂ after plasma treatment, confirming the incorporation of oxygen and the formation of PtSe_xO_y domains near the surface.

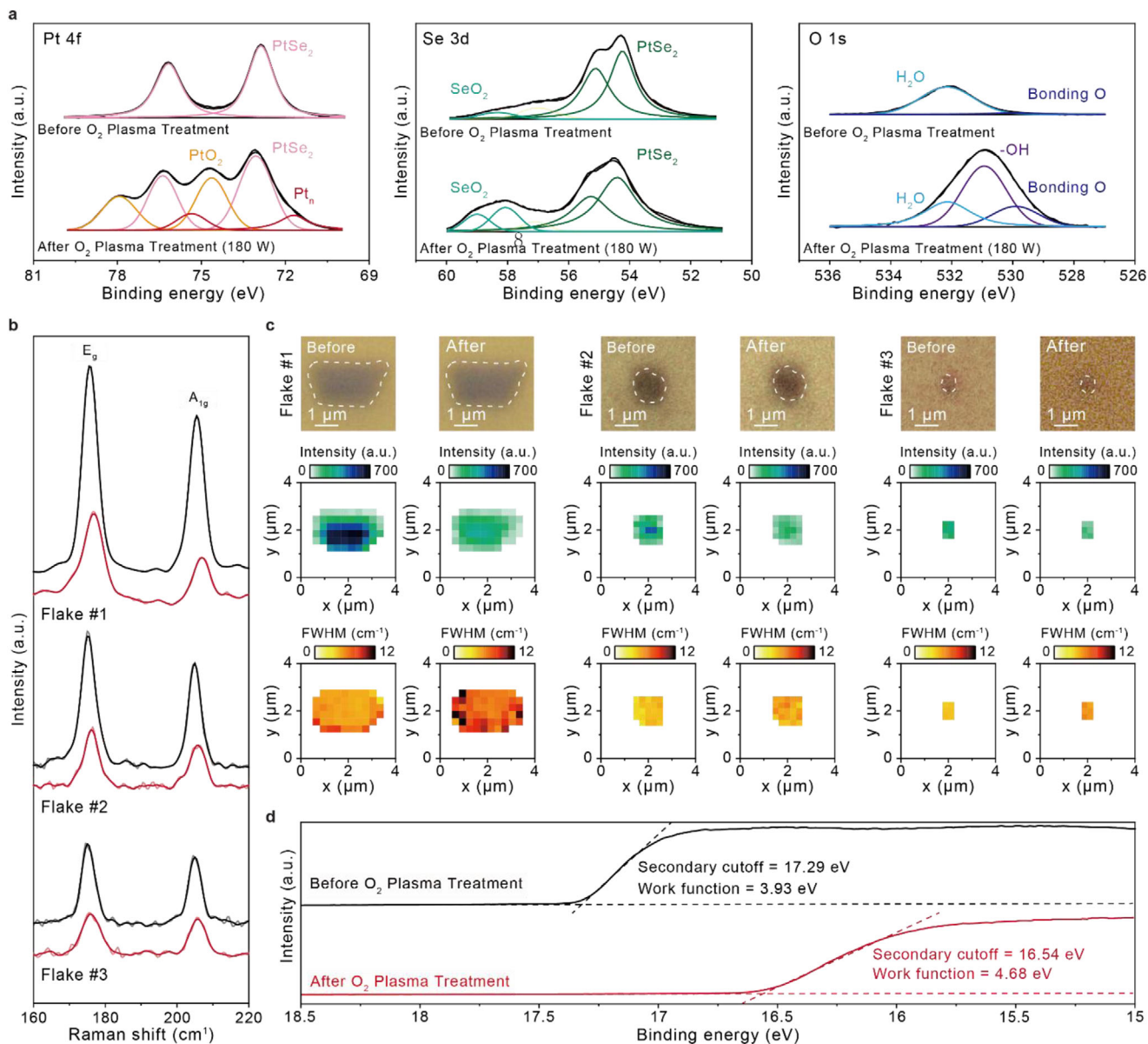


FIGURE 3 | Spectroscopic understanding of PtSe₂ before and after O₂ plasma treatment. (a) XPS spectra of Pt 4f, Se 3d, and O 1s regions for pristine and O₂-plasma-treated PtSe₂, showing the emergence of metallic Pt and oxidized Pt–O/Se–O species. (b) Raman spectra acquired from identical PtSe₂ flakes before and after plasma treatment, extracted from the pixels exhibiting the highest Raman intensity in the corresponding maps. (c) Raman intensity and FWHM maps of the E_g mode for PtSe₂ flakes with different lateral sizes, highlighting spatially heterogeneous surface amorphization after plasma treatment. (d) UPS spectra showing the secondary electron cutoff and the corresponding work-function increase of PtSe₂ after O₂ plasma treatment.

at 58.3 and 59.3 eV, alongside the residual PtSe₂ components at 54.5 and 55.4 eV, confirming partial oxidation of Se sites and the generation of selenium vacancies [54]. The O 1s spectrum further reveals an increased contribution from the lattice oxygen following plasma treatment, consistent with oxygen incorporation into the reconstructed surface [64]. Similar spectral evolutions are observed for lower plasma powers (60 W and 120 W), albeit with reduced intensities (Figures S7–S9), indicating a controllable surface transformation. Together with the structural analyses using HRTEM, these XPS results establish that O₂ plasma treatment drives simultaneous selenium vacancy formation, surface oxidation, and Pt clusterization, yielding a heterogeneous surface composed of crystalline PtSe₂ and amorphous PtSe_xO_y domains, and metallic Pt clusters.

Raman spectroscopy verifies the plasma-induced surface reconstruction while offering spatially resolved phase information at the level of individual PtSe₂ flakes (Figure 3b,c). Unlike XPS, Raman measurements probe single flakes with submicrometer-scale spatial resolution, enabling direct visualization of local structural heterogeneity across each flake. AFM characterizations over representative PtSe₂ flakes with lateral dimensions above, near, and below the average size—determined from the AFM statistics in Figure 1—confirm that the overall flake morphology (except for surface roughness) remains unchanged after plasma treatment. Figure 3b presents representative Raman spectra of the flakes, showing a pronounced weakening of the E_g mode and a concomitant broadening of its full-width-at-half-maximum (FWHM) by approximately 20–40% after O₂ plasma treatment;

these spectral changes signify increased phonon scattering and reduced phonon coherence, indicative of lattice disorder and partial surface amorphization induced by plasma exposure. Furthermore, the extent of these structural modifications scales systematically with the applied plasma power. Comparative Raman analyses of samples treated at 60, 120, and 180 W reveal a progressive decrease in peak intensity and a concurrent increase in FWHM, demonstrating that Raman spectroscopy distinguishes the varying degrees of structural reconstruction (Figure S10). To assess the spatial heterogeneity of these spectral features, Raman mapping was performed across the flakes (Figure 3c). Notably, for the largest flake where Raman mapping provides sufficient spatial resolution, the Raman intensity maps of the E_g mode show a substantial reduction predominantly in the basal-plane regions near the flake center, whereas the corresponding FWHM maps exhibit pronounced broadening toward the flake edges. The intensity attenuation observed in the basal-plane regions suggests the introduction of point defects (e.g., selenium vacancies) eventually forming amorphous PtSe_xO_y domains, which decreases the interaction volume of PtSe_2 as well as screening of the Raman signal due to Pt clusters and amorphous regions on the surface. The FWHM broadening near the edges reflects shortened phonon lifetimes and increased phonon scattering as a result of increased structural heterogeneity, consistent with severe lattice distortion and local amorphization in regions exposed to energetic oxygen plasma species [54, 58, 59, 65].

Ultraviolet photoelectron spectroscopy (UPS) measurements were carried out to probe the electronic structure of O_2 plasma-induced surface reconstruction in layered PtSe_2 (Figure 3d). Similar to XPS, UPS utilizes a global excitation ($\sim 400\text{ }\mu\text{m}$) to probe electronic structure over multiple PtSe_2 flakes within the area of excitation, thereby capturing the net electronic response of the plasma-treated sample surfaces. O_2 plasma treatment induces a pronounced shift of the secondary electron cutoff toward lower binding energy, corresponding to a substantial increase in the work function (Φ) from 3.93 eV for pristine PtSe_2 to 4.68 eV after plasma treatment ($\Delta\Phi \approx 0.75\text{ eV}$). This work-function increase coincides with the emergence of metallic Pt clusters and oxidized surface domains identified by XPS, Raman spectroscopy, and TEM. Given that the intrinsic work function of pristine PtSe_2 ($\approx 4.76\text{--}5.36\text{ eV}$) is lower than that of metallic Pt ($\approx 5.4\text{--}6.1\text{ eV}$), the formation of Pt clusters is expected to contribute to the observed upward shift of Φ [66–69]. In addition, oxygen incorporation at the reconstructed surface further promotes electron depletion, consistent with previous reports on oxygen-induced work-function increase in Pt-based materials [68]. Notably, the increase in Φ reflects a global modification of the surface electronic potential, indicating that the plasma-treated PtSe_2 surface exhibits enhanced electron affinity compared to its pristine counterpart.

To rationalize the formation of the heterostructured catalysts observed by spectroscopy and microscopy, we performed density functional theory (DFT) calculations (Figure 4). Based on the crystallinity and surface reconstruction revealed by HRTEM, the plasma-treated PtSe_2 surface consists of PtSe_xO_y domains accompanied by partial Pt clusterization. Guided by previous studies on interface-modulated, support-stabilized Pt nanoparticles for hydrazine oxidation [29], a theoretical interfacial model was constructed in which a Pt cluster (Pt_n , n is the number of

constituent Pt atoms) is supported on an amorphous PtSe_xO_y layer ($a\text{-PtSe}_x\text{O}_y$), and its stability was systematically compared with that of a Pt cluster supported on crystalline PtSe_2 ($c\text{-PtSe}_2$) (Figure 4a,b, Experimental section). The thermodynamic stability of Pt clusters on different supports was evaluated by calculating the adsorption energy per Pt atom, $[\Delta E_{\text{ads}}(\text{Pt}_n)]/n$ (Figure 4c). The adsorption of Pt clusters on $a\text{-PtSe}_x\text{O}_y$ is significantly more exothermic (-1.03 eV) than on crystalline PtSe_2 (-0.60 eV), indicating that once Pt species are generated during plasma-induced surface reconstruction, they are thermodynamically more strongly stabilized on the amorphous and oxidized surface. Notably, this adsorption energy on $a\text{-PtSe}_x\text{O}_y$ is even more exothermic than the experimental formation enthalpy of PtSe_2 (-0.82 eV) [70], suggesting that the reconstructed $a\text{-PtSe}_x\text{O}_y$ surface provides a deep energetic sink that promotes the retention and aggregation of Pt species. While adsorption energies do not directly describe the reduction or formation energetics of Pt from the parent lattice, these results rationalize the experimentally observed preferential accumulation and stabilization of Pt clusters on $a\text{-PtSe}_x\text{O}_y$ compared with $c\text{-PtSe}_2$, once Pt atoms become available through plasma-induced bond rupture and selenium vacancy formation.

The amorphous nature of PtSe_xO_y domains modulates the surface electronic structures of overall catalysts as evidenced by the UPS results, and these experimental observations are further supported by DFT calculations of the local electrostatic potential and Φ (Figure 4d). A comparison between monolayer $c\text{-PtSe}_2$ and $a\text{-PtSe}_x\text{O}_y$ shows that the amorphous surface exhibits a higher Φ ($+5.91\text{ eV}$) than crystalline PtSe_2 ($+5.48\text{ eV}$), which aligns qualitatively with the trend observed in the UPS measurements showing an increase in work function upon surface oxidation and amorphization [66, 67, 71]. Beyond the monolayer limit, the $a\text{-PtSe}_x\text{O}_y$ domains in the layered system promote strong interfacial charge transfer with the underlying crystalline PtSe_2 bulk (Figure S11); the calculated charge transfer at the $a\text{-PtSe}_x\text{O}_y/c\text{-PtSe}_2$ interface (1.67 e^-) is substantially larger than that at the crystalline PtSe_2 surface (0.09 e^-), indicating strong electronic coupling across the amorphous–crystalline interface. This interfacial charge redistribution further increases the effective work function at the layered system, as reflected by the higher Φ of the trilayer $a\text{-PtSe}_x\text{O}_y/c\text{-PtSe}_2$ model ($+6.08\text{ eV}$) compared to that of trilayer crystalline PtSe_2 ($+4.68\text{ eV}$) (Figure S12). These results indicate that partial surface amorphization into $a\text{-PtSe}_x\text{O}_y$ endows the catalytic surfaces with strong electron affinity, facilitating electron transfer from Pt clusters to the amorphous domains.

Band-decomposed charge density and electronic structure analyses further elucidate how surface amorphization and Pt clusterization cooperatively modulate the near-Fermi electronic states and, consequently, the surface work function relevant to hydrazine oxidation (Figures 4e and S13). For the Pt cluster supported on $c\text{-PtSe}_2$, the electronic states near the Fermi level—including the conduction-band minimum (CBM) and valence-band maximum (VBM)—are dominated by the metallic Pt cluster, with negligible contribution from the PtSe_2 support. This electronic separation indicates very weak Pt–support hybridization near the Fermi level, resulting in limited interfacial charge transfer, as reflected by the small electron accumulation on the Pt cluster (0.27 e^-). Consequently, the crystalline PtSe_2

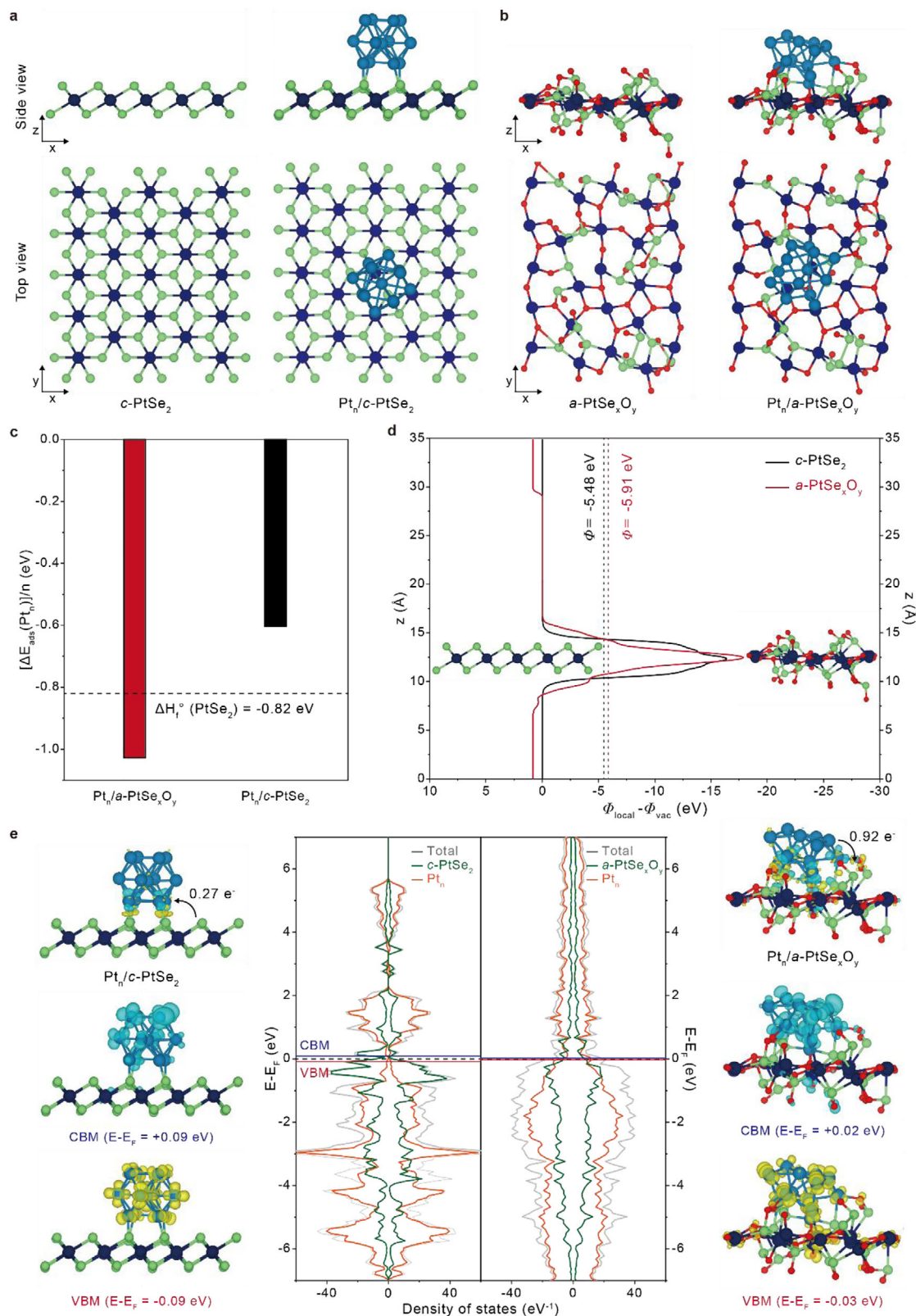


FIGURE 4 | DFT analysis of Pt cluster stabilization and electronic structure modulation. (a, b) Side and top views of the optimized atomic structures of Pt clusters supported on crystalline PtSe_2 ($\text{Pt}_n/\text{c-PtSe}_2$) and amorphous PtSe_xO_y ($\text{Pt}_n/\text{a-PtSe}_x\text{O}_y$), respectively. (c) Adsorption energy per Pt atom, $[\Delta E_{\text{ads}}(\text{Pt}_n)]/n$, for Pt clusters on c-PtSe_2 and $\text{a-PtSe}_x\text{O}_y$, compared with the experimental formation enthalpy of PtSe_2 . (d) Planar-averaged electrostatic potential profiles and calculated work functions for c-PtSe_2 and $\text{a-PtSe}_x\text{O}_y$ -supported systems. (e) Band-decomposed charge density distributions and projected density of states (PDOS) for $\text{Pt}_n/\text{c-PtSe}_2$ and $\text{Pt}_n/\text{a-PtSe}_x\text{O}_y$, illustrating CBM/VBM shifts, near-Fermi-state separation, and enhanced electron transfer at the amorphous interface.

lattice is largely electronically decoupled from the Pt cluster and ineffective in modulating its electron affinity through interfacial coupling. In contrast, when the Pt cluster is supported on α -PtSe_xO_y, the interfacial electronic interaction is substantially enhanced. The electronic states near the Fermi level become spatially hybridized across the Pt cluster and the α -PtSe_xO_y support, enabling more efficient interfacial charge transfer. This enhanced hybridization facilitates electron extraction from the Pt cluster by the amorphous support, increasing the electrophilic and oxidizing character of the Pt active sites and promoting hydrazine adsorption and oxidative activation [72].

The interfacial electronic coupling promotes electron back-donation from the Pt cluster to the support, giving rise to sustained electron depletion at the Pt sites and maintaining their oxidative capability for hydrazine activation under anodic conditions. Consistent with this picture, Bader charge analysis reveals that α -PtSe_xO_y surface extracts a substantially larger amount of charge (0.92 e⁻) from the Pt cluster compared to c -PtSe₂. The resulting electron-deficient Pt clusters are thus stabilized at the α -PtSe_xO_y interface while simultaneously acquiring an electronic structure that is favorable for anodic oxidation reactions [72]. From a catalytic perspective, efficient electronic communication between the Pt cluster and the α -PtSe_xO_y support plays a critical role in HzOR. Strong interfacial hybridization enables effective electron transfer from the Pt cluster to the support under anodic conditions, which facilitates continuous electron withdrawal from adsorbed hydrazine intermediates. This electron-transfer capability suppresses excessive electron accumulation on the Pt cluster, which would otherwise hinder oxidative turnover, and thus sustains efficient catalytic activity.

On the basis of the electronic-structure reconfigurations by surface oxidation and amorphization of layered PtSe₂, we next evaluated whether these theoretical predictions translate into experimentally measurable improvements in HzOR (Figure 5; Figures S14 and S15). Linear sweep voltammetry (LSV) measurements show that the PtSe₂ electrode treated with O₂ plasma at 180 W achieves a current density of 10 mA cm⁻² at a potential of 0.14 V versus RHE, which is significantly lower than that required for pristine PtSe₂ (> 0.6 V vs. RHE) and comparable to that of the benchmark Pt/C catalyst (Figures 5a and S16). We highlight that the HzOR activity on PtSe₂, characterized by a reduced operating potential, gradually improves with increasing plasma power as the amount of Pt clusters increases up to 14.1% (Figure S9). This trend verifies that the formation of Pt clusters accompanied by α -PtSe_xO_y is a key mechanism responsible for the enhanced HzOR activity. Consistent with the LSV results, kinetic analysis based on Tafel plots reveals a pronounced decrease in the Tafel slope from 191 mV dec⁻¹ for pristine PtSe₂ to 72 mV dec⁻¹ after 180 W plasma treatment (Figure S17). The reduced Tafel slope indicates accelerated reaction kinetics and a lower kinetic barrier for the rate-determining step following surface amorphization, oxidation, and Pt clusterization. Notably, the mechanically exfoliated and dry-transferred PtSe₂ flakes provide reliable sample-to-sample uniformity across centimeter-scale electrodes, ensuring batch-to-batch consistency in the measured electrochemical characteristics (Figures 5b and S18). Electrochemical impedance spectroscopy (EIS) measurements further corroborate this kinetic enhancement, revealing a charge-transfer resistance (R_{CT}) of approximately 10 Ω for the plasma-treated

PtSe₂, which is reduced by about 99.7% compared to that of the pristine sample (Figures 5c and S19). The decreased R_{CT} reflects more efficient interfacial charge transfer, in agreement with the increased Φ observed by UPS and the enhanced electron-withdrawing capability predicted by DFT. We note that increased surface hydrophilicity induced by plasma treatment could partially facilitate mass transport (Figure S20), which may contribute to the enhanced HzOR performance.

To estimate the electrochemically active surface area (ECSA), the capacitive current density (Δj) was measured as a function of scan rate (Figures 5d and S21). The double-layer capacitance (C_{DL} , extracted from the slope of the linear fit) increases by approximately 6.4-fold after plasma treatment, indicating a higher density of electrochemically accessible active sites (Figure S21). The larger difference in C_{DL} arises from the use of carbon cloth (CC) as the working electrode substrate for Pt/C, which possesses a substantially higher surface area than the flat Au film supporting PtSe₂ (Figure S21, Experimental section). Importantly, even after normalizing the LSV curves by ECSA, the plasma-treated PtSe₂ maintains a clear advantage over the pristine sample (Figure 5e), confirming that the observed performance enhancement arises not only from increased surface area but also from a genuine improvement in the intrinsic activity of individual active sites. In the context of comparison with benchmark Pt catalysts, this behavior suggests that Pt species in the reconstructed PtSe₂ system are utilized as more efficient active sites, despite the limited fraction of exposed Pt atoms. Building upon this intrinsic activity analysis, the onset overpotential (at low current density of 0.1 mA/cm²) for hydrazine oxidation was further compared before and after ECSA normalization (Figure 5f). The current level at this onset overpotential indicates the intrinsic activity of electrocatalysts under near-equilibrium conditions [8, 13]. On a geometric-area basis, the plasma-treated PtSe₂ at 180 W shows a substantially reduced onset overpotential compared to pristine PtSe₂, indicating that the reaction is initiated at a much lower anodic bias. Notably, this reduced onset overpotential persists after ECSA normalization, demonstrating that the early-stage activation of hydrazine oxidation is intrinsically enhanced at each active site rather than being solely a surface-area-driven effect.

The intrinsic catalytic efficiency was further evaluated by comparative turnover frequency (TOF) analysis (Figure 5g, Experimental section, and Figure S22). The plasma-treated PtSe₂ at 180 W exhibits a markedly higher TOF than all other samples, indicating a substantial enhancement in per-site activity. This TOF increase cannot be attributed solely to the increased number of active sites; instead, it directly correlates with the electronic-structure modulation induced by O₂ plasma treatment. DFT calculations showed that surface amorphization and oxidation induce hybridization of electron-acceptor states within the amorphous PtSe_xO_y domains, which efficiently withdraw electrons from the Pt clusters through strong electronic coupling. This sustained electron depletion suppresses charge accumulation at Pt sites and facilitates successive electron-transfer and deprotonation steps during hydrazine oxidation, giving rise to higher turnover rates. Long-term durability was further assessed by chronoamperometric measurements under continuous operation. The plasma-treated PtSe₂ electrode maintains an almost constant current density over 140 h of continuous operation, whereas the Pt benchmark suffers a severe activity loss of

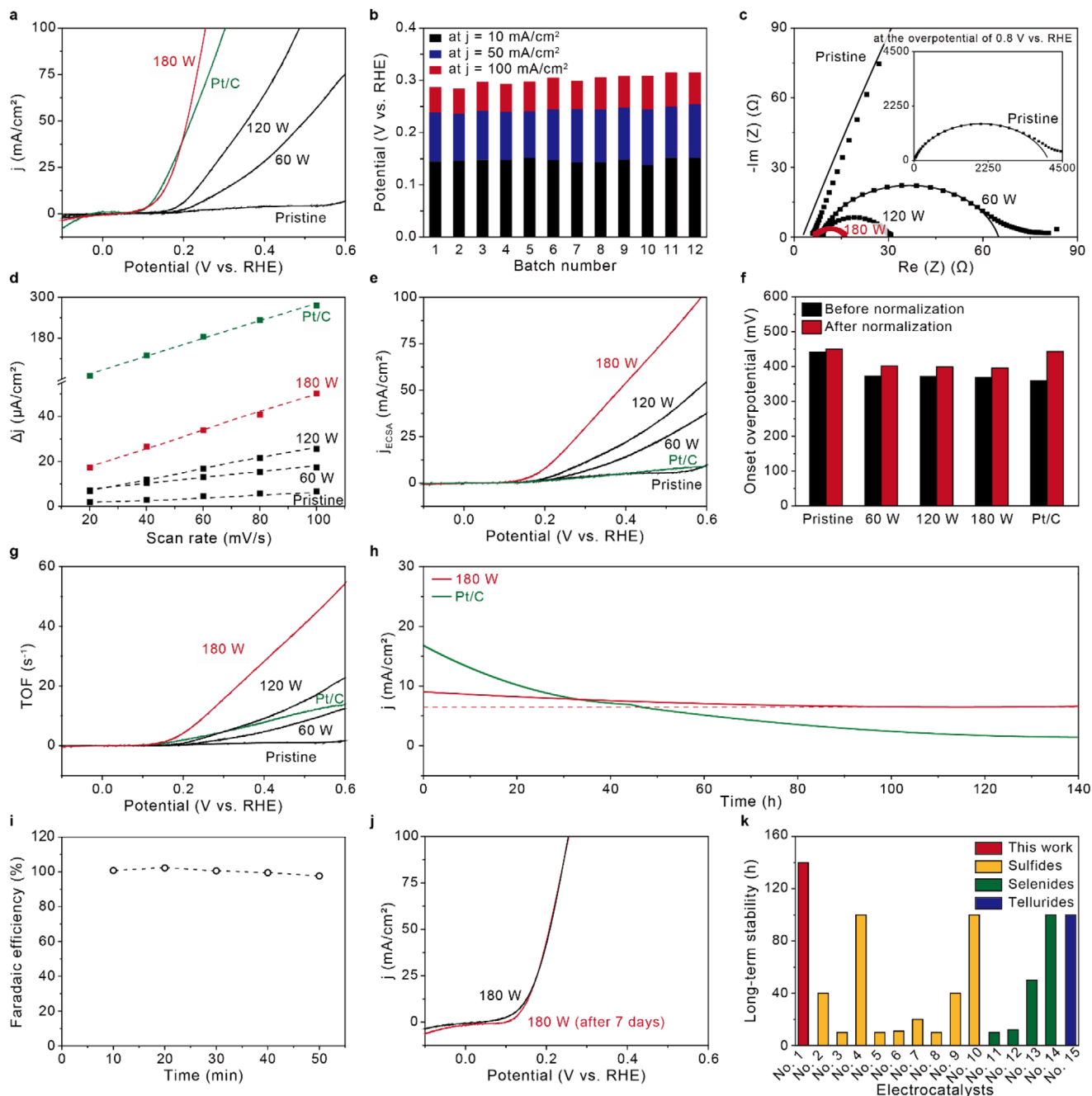


FIGURE 5 | Electrocatalytic HzOR performance of O₂-plasma-treated multilayer PtSe₂. (a) *iR*-corrected linear sweep voltammetry (LSV) curves of pristine and O₂-plasma-treated PtSe₂ (60, 120, and 180 W), compared with a Pt/C benchmark. (b) Batch-to-batch consistency of electrochemical cells, shown as the potentials required to reach current densities of 10, 50, and 100 mA/cm² for each batch. (c) Nyquist plots obtained from electrochemical impedance spectroscopy (EIS), showing reduced charge-transfer resistance after plasma treatment. (d) Capacitive current density as a function of scan rate used to estimate the C_{DL}. (e) LSV curves normalized by the electrochemically active surface area (ECSA). (f) Comparison of onset overpotentials before and after ECSA normalization, highlighting intrinsic activity enhancement. (g) Turnover frequency (TOF) as a function of applied potential for pristine and plasma-treated PtSe₂ and Pt/C. (h) Chronoamperometric stability test conducted at 0.15 V vs. RHE. (i) Faradaic efficiency for N₂ production during the hydrazine oxidation reaction as a function of electrolysis time. For the measurement, the electrochemical cell containing plasma-treated PtSe₂ at 180 W (as a working electrode) was used at a constant potential of 0.6 V vs. RHE. (j) Comparison of alkaline HzOR performance of O₂-plasma-treated PtSe₂ before and after long-term ambient storage, demonstrating the retained electrocatalytic performance over extended storage periods. (k) Comparison of long-term stability between plasma-treated PtSe₂ and transition metal chalcogenide-based electrocatalysts reported in the literature. Detailed information for the catalysts is provided in Figure S28 and Table S2.

~92%, demonstrating the markedly superior electrochemical and structural stability of PtSe₂ under alkaline HzOR conditions (Figure 5h). We confirm that the performance decay of the Pt benchmark originates from intrinsic deactivation of the catalytic sites, rather than physical detachment, as delamination of Pt/C from CC substrate was negligible throughout the measurement (Figure S23).

From a mechanistic perspective, the enhanced stability can be rationalized by the heterostructured architecture, in which electron-withdrawing amorphous domains are electronically coupled to an underlying multilayer *c*-PtSe₂ framework that serves as a robust electron-transport backbone. In contrast to pure Pt systems, where electron accumulation under anodic bias can accelerate surface deactivation, the amorphous–crystalline PtSe₂ platform facilitates continuous electron extraction and transport. This continuous electron withdrawal weakens the binding of nitrogen intermediates and prevents their accumulation (i.e., active-site poisoning), as corroborated by the complete retention of the heterostructure and the absence of nitrogen species in our post-stability XPS analyses (Figure S24). As a result, Pt active sites remain stable during prolonged operation, leading to the superior TOF and the Faradaic efficiency of ~97.7% (Figure 5i). Importantly, this near-unity product selectivity toward N₂ is preserved (~98.6%) even at the lower potential of 0.15 V versus RHE, confirming the consistency and dominance of N₂ generation regardless of the applied potential (Figure S25). Furthermore, rigorous colorimetric quantification confirmed the absolute absence of NH₃ by-products, validating that even when the N–N cleavage pathway is highly active, the intermediates are fully oxidized to N₂ without premature desorption (Figure S26). Moreover, after exposure to ambient conditions for seven days, the catalyst retains nearly identical HzOR activity, highlighting its exceptional resistance to air-induced surface oxidation or deactivation (Figures 5j and S27). Finally, a comparative analysis of key catalytic metrics—including the potential required to reach 10 mA cm^{−2}, Tafel slope, and operational stability—against recently reported transition-metal-chalcogenide-based HzOR catalysts clearly contextualizes the advantages of our system (Figures 5k and S28; Table S2). While several reported catalysts exhibit slightly lower electrode potentials at 10 mA cm^{−2}, our system delivers competitive overall activity combined with notable structural and operational durability. In particular, its continuous operational stability of 140 h represents the highest value among all the state-of-the-art systems compared. These results collectively underscore the effectiveness of our heterostructured catalysts in achieving both relatively high activity and stability for hydrazine electrooxidation.

To quantitatively evaluate the catalytic enhancement by the interfacial modulation, we performed DFT calculations to map the free-energy profiles of hydrazine oxidation and identify the potential-determining step (PDS). (Figures 6a and S29). Hydrazine oxidation can proceed via two distinct pathways: the N–H bond cleavage pathway (NH path) and the N–N bond cleavage pathway (NN path) [22, 73]. Consistent with previous literature, the NH path is thermodynamically more favorable than the NN path at low potentials on both crystalline PtSe₂ (*c*-PtSe₂) and Pt clusters supported on amorphous PtSe_xO_y (Pt_n/*a*-PtSe_xO_y). For both catalysts, the PDS in the NH path corresponds to the initial N–H bond activation of adsorbed hydrazine (*N₂H₄ →

*N₂H₃), with free-energy barriers of +1.00 eV on *c*-PtSe₂ and +0.65 eV on Pt_n/*a*-PtSe_xO_y, indicating a substantially facilitated N–H activation on the interfacial catalyst. The NN path exhibits markedly different energetics depending on the catalyst structure. On Pt_n/*a*-PtSe_xO_y, N–N bond cleavage (NH₂NH₂ → 2NH₂) is strongly stabilized (−1.61 eV), whereas it remains energetically unfavorable on *c*-PtSe₂ (+1.06 eV). Nevertheless, the subsequent N–H bond activation step (*2NH₂ → *2NH) constitutes the highest barrier along the NN path, amounting to +3.00 eV on *c*-PtSe₂ and +1.89 eV on Pt_n/*a*-PtSe_xO_y at U = 0.0 V, where U is a theoretical overpotential per active site. Importantly, increasing the applied potential significantly lowers this barrier on Pt_n/*a*-PtSe_xO_y. At U = 0.6 V versus RHE, the barrier decreases to +0.69 eV, becoming comparable to that of the NH path. The kinetic barriers estimated from relaxed scans of N–H and N–N bond dissociation steps also support that N–N bond dissociation requires higher kinetic barrier (+1.58 eV) than N–H bond dissociation (+0.63 eV) (Figure S30). This difference in kinetic barriers favors N–H dissociation at low potentials, whereas, once the higher barrier for N–N dissociation is overcome at higher potentials, the thermodynamic preference shifts toward N–N dissociation (−1.61 vs. +0.47 eV for N–H dissociation).

In short, these theoretical findings highlight a potential-driven mechanistic transition: this convergence in activation barriers implies that at sufficiently high anodic potentials, the NN path becomes kinetically competitive and can even dominate the reaction pathway on Pt_n/*a*-PtSe_xO_y. These results explain the potential-dependent emergence of *NH₂ intermediates observed from in situ Raman spectroscopy and indicate that Pt_n/*a*-PtSe_xO_y exhibits outstanding catalytic activity along both NH and NN pathways, with the NN path increasingly favored at higher potentials. Raman spectroscopy is sensitive to the molecular polarizability changes of nitrogen-containing bonds, allowing us to effectively and directly capture the vibrational signatures of the surface-bound reaction intermediates in real time [22, 73]. As the applied potential over the three-electrode cell increases, the alternative N–N bond cleavage pathway (NN path) becomes progressively accessible, which is consistent with the in situ Raman observation of *NH₂-related intermediates emerging at higher anodic bias (Figure 6b). The concurrent appearance of N–N vibrational features and the *NH_x signals suggest that N–N bond activation and dissociation are increasingly favored as the potential increases. Notably, at 0.15 V versus RHE—where clear signatures of N–N bond cleavage are observed and where long-term stability tests were conducted—the catalyst maintains markedly superior stability compared with Pt (Figure 5h). This sustained performance under NN-active conditions provides direct evidence that the *a*-PtSe_xO_y layer functions as an effective electron sink, suppressing excessive charge accumulation and preventing the buildup of strongly bound nitrogen intermediates. Furthermore, localized in situ Raman measurements reveal that this robust intermediate evolution is nearly identical at both the central basal plane and the flake edges (Figure S31). This spatial uniformity (relative to the spatial resolution of Raman spectroscopy) highlights the structural advantage of our plasma-treated catalyst: the highly active and poisoning-resistant catalytic mechanism is not confined merely to specific edge sites, but operates uniformly across the entire flake surface, maximizing the effective active area for macroscopic sustained performance.

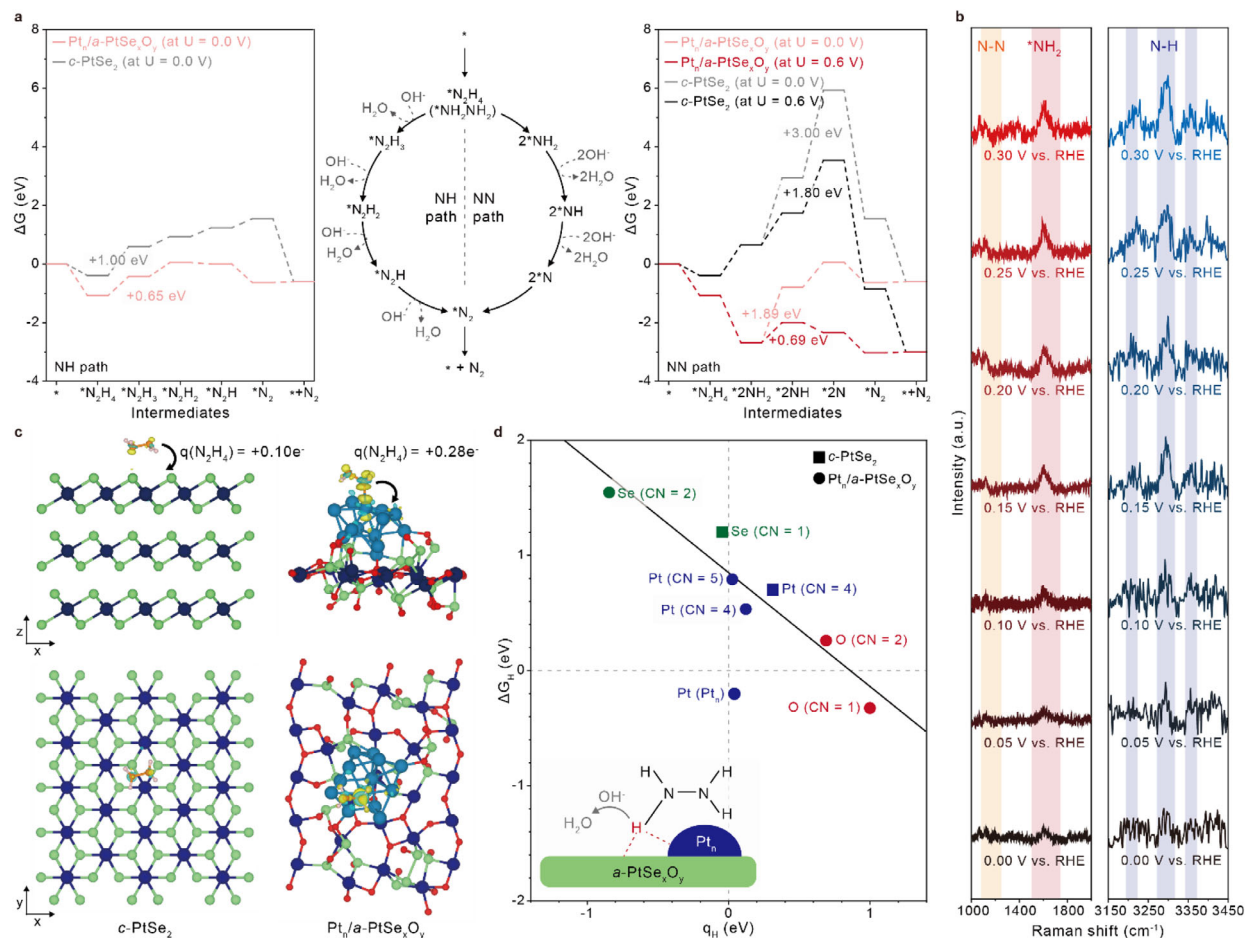


FIGURE 6 | Operando Raman spectroscopy and free-energy analysis of hydrazine oxidation pathways. (a) Gibbs free-energy diagrams for hydrazine oxidation on *c*-PtSe₂ and Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*} along the NH and NN pathways at different applied potentials. (b) In situ Raman spectra collected at increasing applied potentials, showing the emergence of *NH_{*x*} intermediates and potential-dependent activation of the N–N cleavage pathway. (c) Charge density difference plots and Bader charge analysis for *N₂H₄ adsorption on *c*-PtSe₂ and Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*}, illustrating enhanced charge transfer and adsorption strength on the amorphous interface. (d) Correlation between Δ*G*_H and *q*_H at representative surface sites, with a schematic illustration of synergistic N–H activation at Pt clusters and adjacent *a*-PtSe_{*x*}O_{*y*} domains.

To elucidate the electronic origin of the enhanced HzOR activity on Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*}, Bader charge and charge density difference analyses were performed for hydrazine adsorption at the initiation of both the NH and NN pathways (Figure 6c). As discussed above, UPS measurements reveal a pronounced increase in the surface work function upon O₂ plasma treatment, while DFT calculations indicate strong interfacial electronic coupling between the Pt cluster and the *a*-PtSe_{*x*}O_{*y*} support. Together, these results demonstrate that the Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*} ensemble functions as an integrated electronic system with enhanced electron affinity and electron-withdrawing capability. Consistent with the electronic restructuring, adsorbed *N₂H₄ on Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*} exhibits a substantially more positive Bader charge (+0.28 e[−]) compared to that on crystalline PtSe₂ (+0.10 e[−]), reflecting stronger electron transfer from hydrazine to the catalyst surface. This enhanced charge depletion stabilizes hydrazine adsorption and facilitates its oxidative activation into subsequent intermediates. The stronger polarization of *N₂H₄ on Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*} also correlates with the more favorable N–N bond cleavage energetics along the NN pathway. Charge density difference plots further show pronounced electron depletion from the adsorbed N₂H₄ and concurrent electron accumulation within the Pt_{*n*}/*a*-

PtSe_{*x*}O_{*y*} system, confirming its role as an efficient electron sink. In contrast, N₂H₄ adsorption on *c*-PtSe₂ involves only weak charge redistribution, consistent with its lower work function and limited electron-withdrawing capability under anodic bias.

Given that N–H bond cleavage constitutes the PDS in the thermodynamically favored NH pathway, the ability of the catalyst to activate N–H bonds is a critical descriptor of HzOR performance. To assess this capability, the Gibbs free energy of hydrogen adsorption (Δ*G*_H) and the corresponding Bader charge of adsorbed hydrogen (*q*_H) were evaluated at representative surface sites on *c*-PtSe₂ and Pt_{*n*}/*a*-PtSe_{*x*}O_{*y*} (Figure 6d). A clear linear correlation is observed between Δ*G*_H and *q*_H, indicating that sites capable of withdrawing greater electron density stabilize *H more effectively. Notably, oxygen species introduced by O₂ plasma treatment exposes more Brønsted-basic sites with stronger electronegativity (from Se to O) and fewer coordination number (CN) such as Se (CN = 1) and O (CN = 1), which can assist the proton abstraction from N₂H₄. Specifically, an O (CN = 1) site shows even more exergonic stabilization of *H (Δ*G*_H = −0.33 eV) than a Pt site of the Pt cluster (−0.20 eV) in Pt_{*n*}/*a*-

-PtSe_xO_y, and a saturated Pt site (+0.70 eV) in c-PtSe₂. This result demonstrates that the amorphous PtSe_xO_y domains can directly provide favorable proton adsorption sites that synergetically assist proton-coupled electron abstraction from N₂H₄ at the interface of Pt_n/a-PtSe_xO_y.

Thus, the Pt_n/a-PtSe_xO_y system operates as an all-in-one catalytic platform for HzOR, in which electronically activated Pt clusters and the oxidized, unsaturated amorphous matrix cooperatively enable proton-coupled electron abstraction during N–H bond activation. Rather than functioning solely through interfacial charge transfer, the a-PtSe_xO_y matrix plays an active catalytic role by providing Brønsted-basic sites that facilitate proton abstraction from adsorbed hydrazine intermediates, while simultaneously acting as an efficient electron sink. As schematically illustrated in the inset of Figure 6d, electron-deficient Pt clusters generated by charge transfer offer strong adsorption and oxidative activation sites for N₂H₄, whereas the adjacent a-PtSe_xO_y promotes N–H bond cleavage through local proton abstraction (*N₂H₄ → N₂H₃-Pt + H-PtSe_xO_y). The coupled electronic and proton-activation effects in Pt_n/a-PtSe_xO_y accelerate N–H cleavage and stabilize oxidized intermediates, enabling efficient progression along both NH and NN pathways. The cooperative functionality distinguishes the oxidized and unsaturated PtSe_xO_y matrix—formed by O₂ plasma treatment—from purely metallic Pt or conventional Pt-support architectures, as it integrates active sites, electron-sink behavior, and proton-transfer capability within a single catalytic system.

To evaluate the bifunctional electrocatalytic capability of the heterostructures for practical energy-saving water electrolysis, the hydrogen evolution reaction (HER) performance was systematically investigated. As established by previous literature, the formation of atomic- and nanoscale Pt clusters on a PtSe₂ matrix optimizes the local coordination environment and hydrogen binding energetics, thereby boosting HER activity to levels that can even surpass pure Pt [59, 74, 75]. Consistent with these previous experimental and theoretical reports, our plasma-treated PtSe₂ electrodes exhibit substantially enhanced HER activity compared to the pristine counterpart. This is evidenced by significantly lower overpotentials, Tafel slopes, and R_{CT}, ultimately yielding catalytic performance that rivals or exceeds the commercial Pt/C benchmark. (Figure 7a,b; Figures S32 and S33). Notably, the catalytic performance progressively improves as the applied plasma power increases; this power-dependent enhancement indicates that the enhanced HER activity is directly correlated with the increased fractions of Pt clusters on the surfaces. Consistent with the trend observed in our earlier HzOR analysis, the ECSA-normalized curves display a markedly higher intrinsic HER current density for the plasma-treated samples (Figures 7c and S34). Furthermore, to quantitatively validate this maximized atomic utilization efficiency, we evaluated the catalytic performance normalized by the actual Pt mass based on precise ICP-OES measurements (Table S3, Experimental section). Benefiting from the ultralow Pt mass loading of the mechanically exfoliated PtSe₂-based electrode compared to the conventional Pt/C benchmark, our system delivers superior mass-normalized activities for both HzOR and HER (Figures S35 and S36). This confirms that the bifunctional performance is driven by the highly efficient Pt utilization of the dispersed Pt nanoclusters within the heterostructures.

As a proof-of-concept demonstration, we constructed a lab scale two-electrode electrolyzer cell based on plasma-treated PtSe₂ (at 180 W) for overall water splitting (Figure 7d, Experimental section, and Figure S37). This full cell consists of two symmetric electrodes comprising Pt_n/a-PtSe_xO_y (denoted as 180 W || 180 W cell). This cell exhibited a cell potential of 0.19 V at a current density of 10 mA/cm², which is lower than the potential of 0.27 V for a Pt/C || Pt/C cell (i.e., a benchmark system composed of Pt/C as both cathode and anode), indicating that the bifunctionality from the heterostructures is indeed critical for efficient HzOR-assisted water splitting. We observed substantial performance gaps between these two cells, particularly in the high-current-density regime; the 180 W || 180 W cell demonstrated a current density of 100 mA/cm² at a potential of ~0.49 V, while the Pt/C || Pt/C cell delivered only 32% of that current density level (i.e., 32 mA/cm²) at the same potential. To quantitatively investigate the energy-saving advantage of this HzOR-assisted strategy, we further compared our system with the conventional electrolyzer utilizing Pt/C as the cathode (for HER) and iridium oxides (IrO_x) on CC as the anode (for OER). In stark contrast, this Pt/C || IrO_x cell demands substantial cell voltages of 1.56 V and 2.05 V to reach 10 mA/cm² and 100 mA/cm², respectively, inevitably bottlenecked by the sluggish OER kinetics and its high thermodynamic barrier (1.23 V) [7–11]. Remarkably, our device exhibited the long-term stability at the current level of 100 mA/cm² without severe performance degradation for over 140 h of continuous operation (Figures 7e and S38). This durability demonstrates that the aforementioned HzOR stability mechanism—originating from the poisoning-resistant architecture—is preserved and effectively operates even in a practical full-cell system. While the amorphous a-PtSe_xO_y layer acts as an effective electron sink to prevent anodic nitrogen poisoning, the underlying crystalline c-PtSe₂ matrix serves as a robust conductive backbone, ensuring rapid and uninterrupted electron transport across the entire system. Concurrently, the metallic Pt clusters deliver sustained HER activity at the cathode, with their structural and chemical integrity flawlessly retained after prolonged operation (Figure S39).

3 | Conclusion

In summary, we developed a poisoning-resistant, bifunctional Pt_n/a-PtSe_xO_y heterostructures via a single-step plasma treatment of layered PtSe₂ for HzOR-assisted water electrolysis. Mechanistic investigations reveal that the amorphized matrix operates synergistically as an optimal “electron sink” and a proton-accepting environment. This unique architecture renders the embedded Pt clusters electron-deficient while facilitating proton abstraction from nitrogen intermediates, thereby fundamentally suppressing anodic deactivation and delivering significantly increased turnover frequencies during HzOR. Concurrently, the formation of Pt clusters within this optimally modulated coordination environment drastically boosts cathodic HER. Leveraging the enhanced intrinsic per-site activity of both half-reactions over commercial Pt benchmarks, a symmetric two-electrode electrolyzer successfully translated these synergistic bifunctionalities into highly energy-saving overall water splitting. The device achieved exceptional durability, successfully establishing the engineered PtSe₂ system as a robust all-in-one electrocatalytic

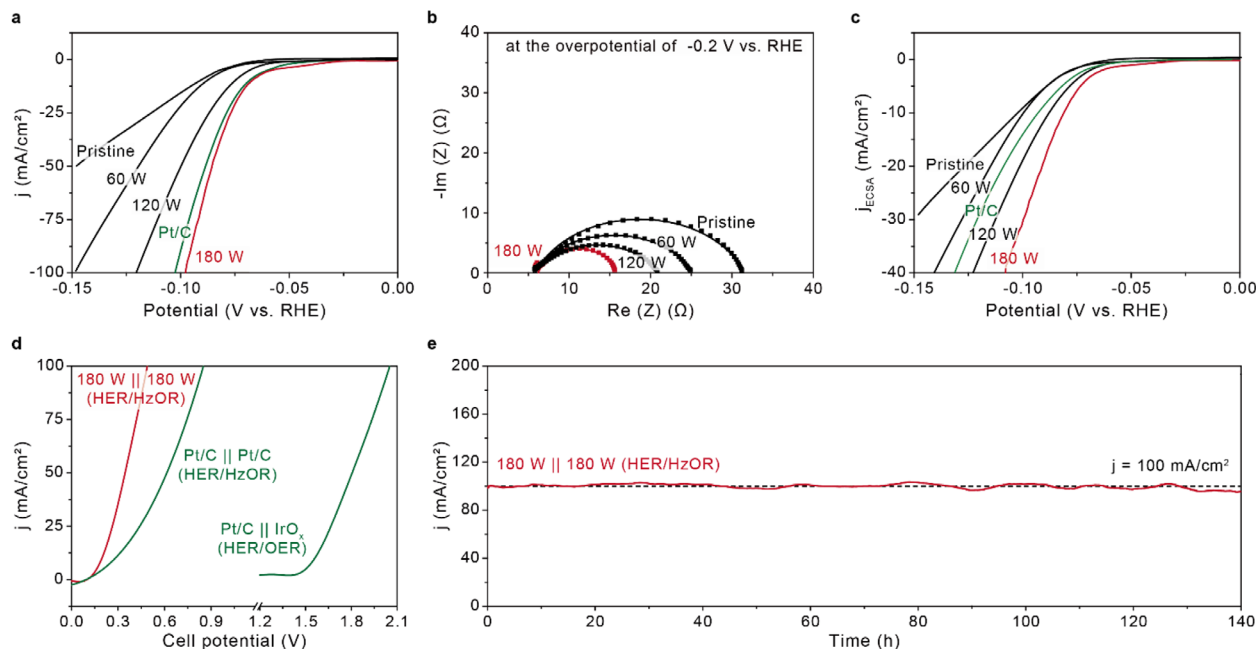


FIGURE 7 | Electrocatalytic HER performance and overall water splitting of the $\text{Pt}_n/\text{a-PtSe}_x\text{O}_y$ heterostructures. (a) iR -corrected linear sweep voltammetry (LSV) curves evaluating the cathodic hydrogen evolution reaction (HER) activity of pristine and O_2 -plasma-treated PtSe_2 (60, 120, and 180 W) in comparison to Pt/C benchmark. (b) Nyquist plots obtained from electrochemical impedance spectroscopy (EIS), demonstrating significantly reduced R_{CT} and accelerated interfacial kinetics upon plasma-induced surface reconstruction. (c) LSV curves normalized by ECSA. (d) I-V curves of the symmetric two-electrode electrolyzer (180 W || 180 W) for hydrazine-assisted overall water splitting, exhibiting substantially lower cell potentials relative to the conventional Pt/C || IrO_x benchmark system. (e) Chronoamperometric durability test of the symmetric electrolyzer continuously operated at a constant cell potential of ~ 0.5 V, highlighting long-term stability up to 140 h. The electrolyte was replaced every 6 h starting 6 h after the initiation of the test.

platform. Compared to other alternative small-molecule oxidation systems [76–78], our work represents significant progress by fundamentally resolving the intrinsic poisoning vulnerability of highly active Pt catalysts. Although future practical deployment requires addressing hydrazine toxicity and minimizing noble-metal usage, ultimately, this architectural design provides a broadly applicable strategy to reduce energy consumption in HzOR-assisted hydrogen production, while establishing a robust paradigm for overcoming intrinsic poisoning challenges in other demanding multi-electron transfer processes, such as ethanol oxidation, nitrogen- and carbon dioxide reduction reactions.

4 | Experimental Section

4.1 | Sample Preparation and Plasma-Induced Reconstruction of PtSe_2

Multilayer PtSe_2 flakes, mechanically exfoliated from bulk crystals, were sequentially transferred onto Au(100 nm)/Cr (15 nm)/ SiO_2/Si substrates using a deterministic stepwise dry transfer method. This procedure ensured highly uniform, large-area surface coverage ($\sim 55\%$), maintaining sample-to-sample consistency. The transferred PtSe_2 electrodes were then subjected to an O_2 plasma treatment (0.25 Torr, 40 sccm, 50 kHz) at systematically varied discharge powers (60, 120, and 180 W) for 10 min to induce surface amorphization, oxidation, and Pt clusterization.

4.2 | Structural, Chemical, and Operando Characterizations

Surface topography, thickness, and roughness evolutions were statistically analyzed using atomic force microscopy (AFM, Bruker MultiMode 8-HR). Atomic-scale structural and elemental characterizations were performed using high-resolution transmission electron microscopy (HRTEM, Titan 80–300, 300 kV) and energy-dispersive x-ray spectroscopy (EDS) following a PMMA-assisted grid transfer. Chemical states, surface compositions, and work functions were determined by x-ray and ultraviolet photoelectron spectroscopy (XPS, Thermo K-Alpha; UPS, Nexsa, He I 21.22 eV). Vibrational properties were probed by micro-Raman spectroscopy (Bruker SENTERRA II, 532 nm). Surface wettability was evaluated via static and dynamic water contact angle measurements. Furthermore, *operando* Raman spectroscopy was conducted in a customized flow cell containing 1.0 M KOH and 0.50 M N_2H_4 under potentiostatic control (from 0.00 to 0.30 V vs. RHE) to monitor the dynamic evolution of surface intermediates during anodic polarization.

4.3 | Electrochemical Measurements (Half-Cell and Full-Cell)

Electrochemical evaluations were performed using a ZIVE SP1 potentiostat in a standard three-electrode configuration. The HzOR performance was measured in N_2 -saturated 1.0 M KOH containing 0.50 M N_2H_4 using a Hg/HgO reference electrode,

whereas the HER performance was evaluated in 0.50 M H₂SO₄ using an Ag/AgCl reference electrode. Linear sweep voltammetry (LSV) was recorded with real-time dynamic *i*R correction. The electrochemically active surface area (ECSA) was estimated from the electrochemical double-layer capacitance (*C*_{DL}) derived from scan-rate-dependent cyclic voltammetry (CV). Turnover frequencies (TOF) were calculated based on the quantified number of active sites from full CV integrations. Faradaic efficiencies for gaseous products (N₂) were verified using online gas chromatography (GC, YL ChroZen) equipped with TCD and FID. For practical demonstration, a lab-scale symmetric two-electrode electrolyzer was constructed using the engineered PtSe₂ sample as both the anode and cathode in an H-cell separated by a bipolar membrane. For comparison, conventional benchmark cells comprising Pt/C and IrO_x/CC electrodes were also evaluated.

4.4 | Density Functional Theory (DFT) Calculations

Spin-polarized DFT calculations were performed using VASP 5.4.1. The Perdew-Burke-Ernzerhof (PBE) functional with Grimme's D3 van der Waals correction was employed. The plane-wave cutoff energy was set to 400 eV. The PtSe_xO_y amorphous structure was modeled via a "melt-and-quench" molecular dynamics approach (510 to 300 K), and a Pt₁₃ icosahedral cluster was incorporated to represent the Pt active sites. Gibbs free energies for the intermediate states were calculated at 298.15 K, and interfacial electronic structures were quantitatively analyzed using Bader charge analysis.

4.5 | Extended Experimental Methods

To maintain the concise nature of the main text while providing full transparency, comprehensive experimental procedures and instrumental parameters are detailed in the Supporting Information (SI). These extended methods include the exact protocols for the deterministic stepwise dry transfer of PtSe₂, the PMMA-assisted grid transfer technique for HRTEM, and the specific customized configuration for the *operando* electrochemical flow cell. Furthermore, the exact ink formulation recipes for commercial benchmark catalysts (Pt/C and IrO_x/CC), gas chromatography (GC) calibration procedures for Faradaic efficiency calculations, and extended computational parameters for the DFT modeling are fully described in the SI.

Author Contributions

W.K.L. developed the concept and designed the experiments. S.K.H., H.J., and J.H.Y. fabricated samples and prepared the electrochemical cells to test HzOR performance. S.K.H., H.J., and S.J.B. characterized the samples using AFM. I.K. performed gas chromatography and indophenol blue analyses. S.K.H., H.J., J.H.Y., and J.H. measured UPS and Raman spectroscopy. S.K.H. and H.J. conducted in situ Raman analysis. S.K.H., H.J., J.L., and S.S. measured XPS spectroscopy. S.K.H., H.J., J.H.Y., M.K.C., and I.S.K. characterized the samples using TEM. Y.K. and M.K. developed theoretical models. S.K.H., H.J., J.H.C., S.Y.K., Y.J.J., I.S.K., M.K., and W.K.L. analyzed experimental and theoretical data. W.K.L. supervised the experiments. S.K.H., Y.K., H.J., S.Y.K., Y.J.J., I.S.K., M.K., and W.K.L. wrote manuscripts.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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Supporting Information

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